

Adı: Furkan ~~SA~~

Soyadı: SADAN

No: 081202056

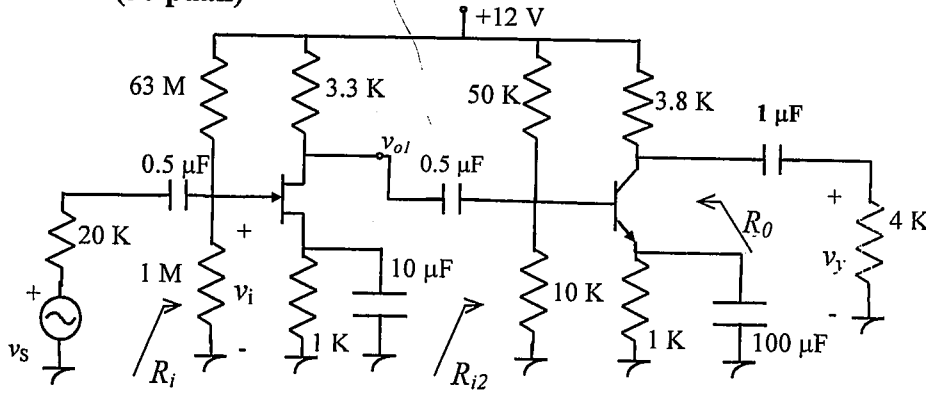
Elektronik II Final Sınav Soruları

03.06.2011

3.
19

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S.1- (30 puan)



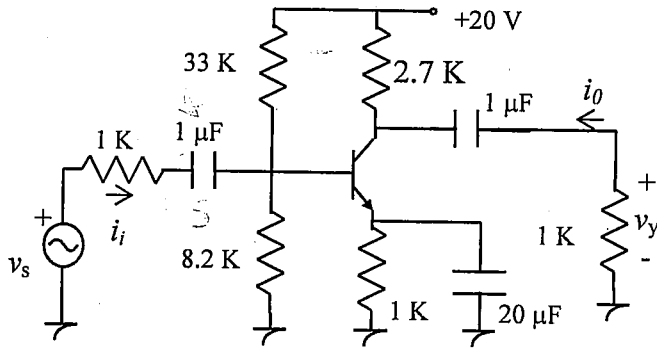
$$g_m = 2.16 \text{ mS}, \quad h_{fe} = \beta = 150, \\ h_{ie} = 2 \text{ k}\Omega$$

$$\text{a) } R_i = ?, R_{i2} = ? \text{ ve } R_o = ?$$

$$\text{b) } A_{v1} = \frac{v_{o1}}{v_i} = ? \quad A_{v2} = \frac{v_o}{v_{o1}} = ?$$

$$A_{vs} = \frac{v_o}{v_s} = ?$$

S.2- (20 puan)

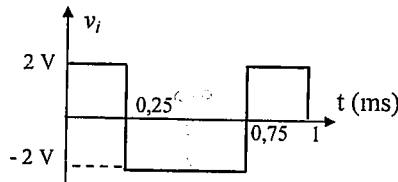
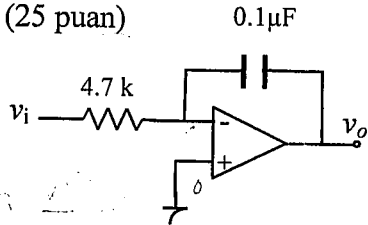


$$h_{fe} = 150, \quad h_{ie} = 1.3 \text{ k}\Omega, \quad C_{bc} = 10 \text{ pF}, \\ C_{ce} = C_{be} = 2 \text{ pF}, \quad C_{w1} = C_{w2} = 3 \text{ pF} \text{ ve} \\ C_1 = C_{bc}(1 - A_{vs}) \text{ ve } C_2 = C_{bc}(1 + 1/A_{vs}) \text{ olmak üzere,}$$

$$\text{a) } A_{vs} = \frac{v_o}{v_s} = ? \quad \text{b) alt kesim frekansı} = ?$$

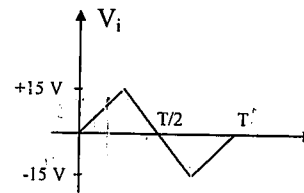
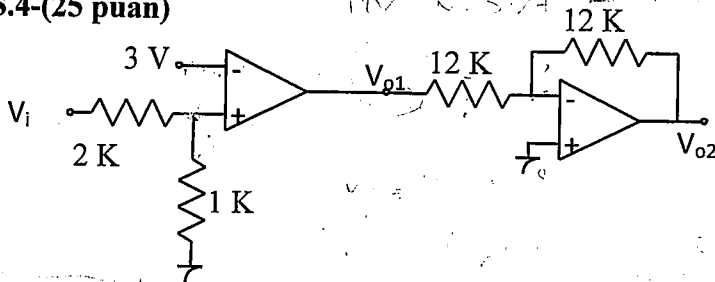
$$\text{c) üst kesim frekansı} = ?$$

S.3- (25 puan)



v_o sinyalini 0-1 ms aralığında ölçekli olarak çiziniz. ($\pm V_{cc} = \pm 12 \text{ V}$)

S.4- (25 puan)



Her iki opamp için $\pm V_{cc} = \pm 12 \text{ V}$ olmak üzere, V_{o1} ve V_{o2} sinyallerini ölçekli olarak çiziniz.

Not: Sınav süresi 90 dakikadır. Başarılar dileriz.

Prof. Dr. Saadetdin HERDEM

Dr. Seral ÖZŞEN